DfX Rationale – Team 05

1. Concept Selection Matrix (Pugh Matrix)

This section documents the selection of the primary control strategy for the line-follower robot's digital twin. Three feasible alternatives were evaluated against key design criteria derived from the requirements document and system architecture.

Design Alternatives Evaluated:

1. **Alternative 1: PID Controller:** A classical control loop using Proportional, Integral and Derivative gains, acting on the lateral error (e_line). This controller resides in the Controller block, receives e_line from the Sensors block and outputs u_L/u_R to the Actuators block.
2. **Alternative 2: State-Space Controller (LQR):** A model-based controller using full-state feedback. This requires a plant model (derived from the Plant block dynamics) and aims to optimize performance. It also resides in the Controller block.
3. **Alternative 3: Cascade PID:** A hierarchical control approach where an inner loop controls motor speed (v, omega) and an outer loop controls lateral position (e_line). This adds complexity but can improve robustness to disturbances within the Plant and Actuators blocks.

Selection Criteria and Weights:

Criterion	Weight	Justification
Performance (IAE)	40%	This is the primary functional requirement (R-TRACK-02: IAE < 0.030 m·s). The controller's ability to minimize tracking error is paramount.
Tuning Ease	25%	The CA2 timeline requires a quick and successful implementation. A controller that is easier to tune reduces project risk.
Implementation Complexity	20%	As a digital twin implementation, the code must be reliable and real-time feasible. Lower complexity reduces the chance of bugs and simplifies integration with the Plant and Sensors blocks.
Robustness to Disturbances	15%	The system must handle disturbances like sharp curves (S1 scenario) and sensor noise. The selected controller should be robust enough to maintain stable tracking.

Pugh Matrix & Architecture Linkage:

The evaluation is based on the system's high-level architecture, as defined in the Block Definition Diagram (BDD) and Internal Block Diagram (IBD). The table below scores each alternative against the established criteria, considering how each controller interacts with the core architectural blocks (Controller, Plant, Sensors, Actuators).

Criterion	**Weight**	**PID Controller**	**State-Space (LQR)**	**Cascade PID**
Performance (IAE)	0.4	**8** (Good tracking with proper tuning)	**9** (Optimal performance possible)	**8** (Potentially better than single PID)
Tuning Ease	0.25	**9** (Ziegler-Nichols method is straightforward)	**6** (Complex, requires a validated plant model)	**6** (Multiple loops require careful coordination)
Implementation Complexity	0.2	**9** (Simple code, direct e_line -> u_L/u_R mapping)	**7** (Requires matrix operations and a robust model)	**7** (More complex than single PID)
Robustness to Disturbances	0.15	**8** (Proven robustness to model inaccuracies)	**7** (Sensitive to model mismatch)	**8** (Inner loop can reject speed disturbances)
Weighted Score	1	**8.35**	**7.55**	**7.4**

Scoring Scale: 1-10, where 10 is the most favorable.

Weight Justification:

- **Performance (40%):** The highest weight is assigned to performance as it is the direct measure of the system's primary function: accurate line tracking (IAE < 0.030 m·s).
- **Tuning Ease (25%):** With a tight deadline for CA2 (4 March), the ability to quickly calibrate the controller to meet specifications is critical for project success.
- **Implementation Complexity (20%):** This ensures the controller code can be reliably integrated into the digital twin. Lower complexity translates to fewer potential bugs and easier debugging.
- **Robustness (15%):** While important, this is weighted slightly lower than immediate performance and timeline constraints for the SDR phase.

Conclusion:

The PID Controller was selected as the primary control strategy for CA2 and will be implemented for the S1 verification. It achieved the highest weighted score (8.35), offering the best balance of strong performance and implementation simplicity. The PID's straightforward tuning process aligns well with the CA2 timeline, making it the low-risk choice for achieving the initial performance targets. The State-Space controller, while offering potential performance benefits, scored lower (7.55) due to its higher tuning complexity and reliance on an accurate plant model, which is not yet fully validated. The Cascade PID, while robust, scored similarly (7.40) due to its increased complexity.

2. Life Cycle Assessment (LCA) Screening

A qualitative LCA screening was performed to identify the most significant environmental impacts associated with the line-follower robot's lifecycle. This analysis considers the embodied and operational impacts, guiding future design for sustainability.

Top 3 Environmental Hotspots:

Based on an analysis of the system's Bill of Materials and operational profile, the following components represent the primary environmental hotspots:

Rank	Component	Lifecycle Stage	Environmental Impact	Severity	Mitigation Strategy
1	**Alkaline Batteries**	Disposal (End-of-Life)	Alkaline batteries are classified as hazardous waste due to their heavy metal content (zinc, manganese). Improper disposal leads to soil and water contamination. While some are recyclable, a significant portion ends up in landfills.	**MEDIUM**	Use rechargeable NiMH batteries to reduce waste. Ensure proper battery recycling through designated e-waste programs. In future designs, consider a built-in rechargeable Lithium-Ion or LFP battery pack to eliminate single-use waste.
2	**DC Motors**	Operation (Use Phase)	The motors, with a baseline efficiency of ~75%, convert 25% of their electrical energy into waste heat. Over the robot's 75+ second operational scenarios, this represents a significant energy loss, contributing to overall energy consumption and faster battery depletion.	**MEDIUM**	For future designs, select motors with a rated efficiency of **≥85%**. This would reduce energy waste by over 30%, improve system efficiency, and extend battery life, reducing the frequency of battery changes.

3	**ABS Plastic Chassis**	Manufacturing & End-of-Life	ABS (Acrylonitrile Butadiene Styrene) is a petroleum-based plastic that is not readily biodegradable. While the material mass is small, its production is energy-intensive and its disposal contributes to long-term plastic pollution.	**LOW-MEDIUM**	When sourcing or 3D printing the chassis, consider sustainable alternatives like **PLA (Polylactic Acid)** , a bio-based plastic, or **recycled PETG**, which gives a second life to plastic waste.

Mitigation Considerations for Future Work:

- **Battery:** Transition to rechargeable NiMH batteries or a built-in Lithium-Ion/LFP pack to eliminate single-use alkaline waste. If alkaline batteries must be used, implement a team policy for proper recycling.
- **Motors:** Prioritize high-efficiency (≥85%) motors in the bill of materials to minimize operational energy losses and extend battery runtime.
- **Chassis:** Adopt sustainable materials like PLA or recycled PETG for prototyping and manufacturing to reduce reliance on fossil fuels and improve end-of-life options.